

Hongshan Guo

Assistant Professor / Department of Architecture
The University of Hong Kong

hongshan@hku.hk / thermalcomfort.zone / +852 9677 8596

[ORCID 0000-0002-8716-0343](#) / [Publications](#)

I study how people, building systems, and climate interact to shape thermal protection and energy demand. My work asks how building decisions can account for the likelihood and consequences of unacceptable discomfort across diverse occupants and uncertain climate futures.

01 / RESEARCH PROGRAMME

Selected work links to the bibliography below.

People / Thermal protection and human variability

Predicting how hot or cold people feel, accounting for human heat generation and context.
human heat defaults [J10]; contextual drivers [J07]; thermal sensation probabilities [J06].

Systems / Integration and energy uncertainty

Linking human response and building control to assess discomfort risk alongside energy demand.
co-simulation [J09]; model benchmarking [A02]; HVAC supervision using sensation probabilities [A01].

Climate / Future weather and resilience

Future weather, input fidelity, and uncertainty in climate-resilient building performance.
sAMY — generation [J03] and input fidelity [J05]; cooling and ageing [A03].

02 / APPOINTMENTS

2023–present	Assistant Professor / The University of Hong Kong Department of Architecture; appointed September 2023.
2020–2023	Principal Data Scientist / Bank of New York Mellon Led ML deployment and model governance for daily USD 29–43 billion Federal Reserve account-balance forecasts.
2019–2020	Postdoctoral Researcher / Princeton University Building science, sensing, and environmental modeling.

03 / EDUCATION

2019	Ph.D., Architecture / Princeton University Architectural Technology & Computational Design.
2013	M.Eng., Mechanical Engineering / Columbia University
2012	B.Eng., Architectural Engineering / Harbin Institute of Technology HVAC Engineering.

04 / RESEARCH FUNDING

2026–	ASHRAE Research Project 1959-TRP / \$55,000 Principal investigator; demographic-aware metabolic-rate defaults for thermal comfort standards. Awarded February 2026; 15 months from April 2026. Sponsor: TC 4.1.
-------	---

05 / TEACHING INNOVATION / FUNDING

- 2025–2028 **Socratic Oracle / HK\$1,000,000**
Principal investigator, Teaching Development & Language Enhancement Grant, HKU. Multimodal voice-AI tutor for evidence-based learning.
- 2026–2028 **AI Design Coach / HK\$300,000**
Principal investigator, Teaching Development Grant, HKU. Conversational support for design narratives across Architecture, Geography, and Computer Science.

06 / HONORS & AWARDS

- May 2026 **Best Paper Award / IAQVEC 2026, Los Angeles**
Co-recipient with PhD student Kanxuan He, as supervisor and co-author, for [climate-zone classification for building codes](#).
- 2025 **1st Place / ICML CO-BUILD Smart Building Competition**
Faculty mentor and author.
- 2025 **3rd Place / NeurIPS Urban AI Workshop, Buildings Challenge**
Faculty mentor.
- 2021 **Gartner Eye on Innovation Award / BNY Mellon**
- 2018 **Lowry Methodology Award / International Conference of Urban Climate**

07 / SUPERVISION

- 2025–present **Kanxuan He / PhD, HKU**
Probabilistic control and climate resilience.
- 2024–present **Yu Chang / PhD, HKU**
Cross-cultural thermal comfort; first-author review in *Renewable and Sustainable Energy Reviews* (2025).
- 2026 **MArch thesis supervision / HKU**
Physics-informed occupant modeling; EEG-based neuroarchitecture.

08 / TEACHING

- 2025–present **ARCH7476 / Evidence-Based Generative Design**
Designed and taught; MArch / UG / PhD. Reproducible computational design.
- 2024–present **CCGL9065 / Our Response to Climate Change**
Designed and taught; undergraduate. Climate action and public exhibition *Hong Kong 2100*.
- 2023–present **DESN2003 / Research for Innovation**
Designed and taught; undergraduate. Research methods and communication.
- 2015–2018 **Prior teaching / Princeton University**
Mini-courses on IoT sensing and thermodynamics laboratories.

09 / SERVICE & PUBLIC SCHOLARSHIP

Academic leadership. Co-Chair, Technical Development & Proceedings Integration, CAAD Futures 2025.

Peer review. Journals including *Energy and Buildings*, *Building and Environment*, and *Applied Energy*; workshops at CVPR, ICML, and NeurIPS.

Exhibition. Co-author and exhibitor, *Social Condenser Extraordinaire: Hong Kong's Municipal Services Buildings*, *Projecting Future Heritage: A Hong Kong Archive*; Hong Kong collateral exhibition, 19th Venice Architecture Biennale, May–November 2025.

10 / PUBLICATIONS

Bold reference numbers mark selected work and link back to the research programme. [DOI] opens the article.

Journal articles / accepted and in press

- A01 Guo H.** (2026). Delivered-Energy and Operative-Temperature Trade-offs for Learned-Probability-Informed HVAC Supervision in a Weather Stress Grid. *Energy and Buildings*. **Accepted 27 September 2026**.
- A02 Guo H.,** He K. (2026). Lower Peaks, Similar Growth: Isolating the Value and Limits of AI-Driven Thermal Sensation Modeling for Building–Grid Resilience Under Future Climates. *Engineering Applications of Artificial Intelligence*. **Accepted 15 September 2026**.
- A03 Guo H.** (2026). Equal cooling setpoints, unequal thermal outcomes: Residential cooling service under future weather and ageing. *Energy*, article 142403. **In press**. [DOI].

Journal articles / published

- J01 Chang Y., Guo H.** (2026). Temporal regime diagnostics of dining-space CO₂ under occupancy loading: A long-horizon operational study with dense-grid seed evidence. *Building and Environment*, **305**, 115257. Published online 14 September 2026. [DOI].
- J02 Guo H.,** Anselmo S., Ferrara M., Niu S., Chi B., Wang X. (2026). Automated urban energy assessment: From thermal flyover to AI-driven retrofit prioritization for sustainable cities. *Sustainable Cities and Society*, **148**, 107589. [DOI].
- J03 Guo H.,** He K. (2026). Scenario-conditioned actual meteorological years (sAMY): A stochastic weather generator using multi-decadal observations. *Energy and Buildings*, **364**, 117508. [DOI].
- J04 Cen C., Guo H.,** Chew L.W., Wong N.H. (2026). Experimental study on gender differences in thermal comfort and physiological responses in fan-assisted cooling environments. *Energy and Buildings*, **360**, 117365. [DOI].
- J05 Guo H.,** He K. (2026). Input quality, not statistical complexity, determines climate-adapted weather file fidelity: A causal decomposition of degree-day errors. *Energy*, **353**, 140867. [DOI].
- J06 Guo H.,** Aviv D. (2026). From seven points to probabilities: Ordinal learning for risk-aware thermal comfort prediction. *Building and Environment*, **295**, 114426. [DOI].
- J07 Guo H.,** Pigliautile I., Chang Y., Qiao Q., Li Y. (2026). Toward smarter HVAC control: Machine learning reveals hidden drivers in thermal comfort databases. *Energy and AI*, **24**, 100709. [DOI].
- J08 Guo H.,** Chang Y., Li Y., Zhou Y., Qiao Q., Lai C.Y., Schuldenfrei E. (2026). Ventilation–energy trade-offs in retrofitted Hong Kong wet markets. *Energy and Buildings*, **355**, 116918. [DOI].
- J09 Guo H.,** He K., Xu Y., Lei Y. (2026). A co-simulation methodology for integrating data-driven thermal sensation models with building energy control. *Energy and Buildings*, **353**, 116745. [DOI].
- J10 Guo H.,** Sun R., Xu Y. (2025). Correcting the 120-watt assumption: Demographic-aware metabolic rates for energy savings and thermal comfort equity in buildings. *Energy and Buildings*, **349**, 116525. [DOI].
- J11 Chang Y., Guo H.,** Li Y., Pigliautile I., Chi B. (2025). A data-driven qualitative review of thermal comfort studies: Bridging the gap between western and eastern perspectives. *Renewable and Sustainable Energy Reviews*, **224**, 116020. [DOI].
- J12 Guo H.,** He K., Luo Y., Chang Y. (2025). Physics-Informed Neural Networks for robust thermal comfort prediction: Overcoming data quality limitations through physiological constraints. *Building and Environment*, **285**, 113588. [DOI].

- J13 Aviv D., **Guo H.**, Middel A., Meggers F. (2021). Evaluating radiant heat in an outdoor urban environment: Resolving spatial and temporal variations with two sensing platforms and data-driven simulation. *Urban Climate*, **35**, 100745. [DOI].
- J14 Ma N., Aviv D., **Guo H.**, Braham W.W. (2021). Measuring the right factors: A review of variables and models for thermal comfort and indoor air quality. *Renewable and Sustainable Energy Reviews*, **135**, 110436. [DOI].
- J15 **Guo H.**, Ferrara M., Coleman J., Loyola M., Meggers F. (2020). Air temperature and mean radiant temperature data, collected and simulated across a radiantly-heated high-bay laboratory. *Data in Brief*, **30**, 105192. [DOI].
- J16 Teitelbaum E., Chen K.W., Meggers F., **Guo H.**, Houchois N., Pantelic J., Rysanek A. (2020). Globe thermometer free convection error potentials. *Scientific Reports*, **10**(1), 2652. [DOI].
- J17 **Guo H.**, Ferrara M., Coleman J., Loyola M., Meggers F. (2020). Simulation and measurement of air temperatures and mean radiant temperatures in a radiantly heated indoor space. *Energy*, **193**, 116369. [DOI].
- J18 **Guo H.**, Aviv D., Loyola M., Teitelbaum E., Houchois N., Meggers F. (2020). On the understanding of the mean radiant temperature within both the indoor and outdoor environment, a critical review. *Renewable and Sustainable Energy Reviews*, **117**, 109207. [DOI].
- J19 Luo Y., **Guo H.**, Meggers F., Zhang L. (2019). Deep coaxial borehole heat exchanger: Analytical modeling and thermal analysis. *Energy*, **185**, 1298–1313. [DOI].
- J20 **Guo H.**, Luo Y., Meggers F., Simonetti M. (2019). Human body exergy consumption models' evaluation and their sensitivities towards different environmental conditions. *Energy*, **183**, 1075–1088. [DOI].
- J21 Luo Y., Zhang L., Bozlar M., Liu Z., **Guo H.**, Meggers F. (2019). Active building envelope systems toward renewable and sustainable energy. *Renewable and Sustainable Energy Reviews*, **104**, 470–491. [DOI].
- J22 **Guo H.**, Teitelbaum E., Houchois N., Bozlar M., Meggers F. (2018). Revisiting the use of globe thermometers to estimate radiant temperature in studies of heating and ventilation. *Energy and Buildings*, **180**, 83–94. [DOI].
- J23 Meggers F., **Guo H.**, Teitelbaum E., Aschwanden G., Read J., Houchois N., Pantelic J., Calabrò E. (2017). The Thermoheliodome – “Air conditioning” without conditioning the air, using radiant cooling and indirect evaporation. *Energy and Buildings*, **157**, 11–19. [DOI].
- J24 Meggers F., Aschwanden G., Teitelbaum E., **Guo H.**, Salazar L., Bruelisauer M. (2016). Urban cooling primary energy reduction potential: System losses caused by microclimates. *Sustainable Cities and Society*, **27**, 315–323. [DOI].

Conference papers and workshop contributions

- C01 He K., **Guo H.** (2026). Data-driven cross-regional climate zone classification for building codes: A future-compatible framework. *E3S Web of Conferences*, **716**, 10011. **Best Paper Award, IAQVEC 2026 (May)**. [DOI].
- C02 He K., **Guo H.** (2025). A temporal features-enhanced mixture-of-experts approach for indoor temperature prediction. *ICML CO-BUILD Workshop*, oral presentation. **1st Place, Smart Building Competition**.
- C03 Niu S., **Guo H.**, Ferrara M., Anselmo S. (2025). Thermal-SAM: Adversarial prompt-based unsupervised building segmentation in thermal aerial imagery. *ICML CO-BUILD Workshop*, poster. **3rd Place, NeurIPS Urban AI Workshop (Buildings Challenge)**.

- c04 **Guo H.**, Chang Y., Hu D. (2025). ML-driven sensitivity analysis for lean HVAC: New insights from large-scale comfort data. *ICML CO-BUILD Workshop*, poster.
- c05 **Guo H.**, Zhou Y., Lai C.Y., Ren C. (2025). From open air to air-tight: Analyzing the ventilation overhaul in Hong Kong's wet markets and its implications. In *Multiphysics and Multiscale Building Physics* (IBPC 2024), LNCE **555**, pp. 55–60. Springer; online December 2024. [DOI].
- c06 **Guo H.**, Coleman J., Gullapalli I. (2025). Accelerating NZEB design optimization through LLM-based standardization and compliance checking. In *Multiphysics and Multiscale Building Physics* (IBPC 2024), LNCE **553**, pp. 585–591. Springer; online December 2024. [DOI].
- c07 **Guo H.**, Meggers F. (2019). Charging and discharging a coaxial borehole heat exchanger as a battery. *Proceedings of Building Simulation 2019*, pp. 1749–1754. [DOI].
- c08 **Guo H.**, Teitelbaum E., Meggers F. (2019). Humidifying without adding humidity: Psychrometric shifts in humidity from air temperature setbacks enabled by radiant heating or cooling. *Proceedings of Building Simulation 2019*, pp. 2562–2568.
- c09 **Guo H.** (2018). Analytical evaluation of thermal comfort implications from an expanded range of air temperatures and velocities. *15th International Conference on Indoor Air Quality and Climate (Indoor Air 2018)*.
- c10 Coleman J., Teitelbaum E., **Guo H.**, Read J., Meggers F. (2017). Examining Architectural Air and Temperature with Novel Sensing Techniques. *Energy Procedia*, **122**, 1135–1140. [DOI].
- c11 **Guo H.**, Meggers F. (2017). Geothermal distribution network modeled as Heat Exchanger Network to be optimized with Mixed Integer Nonlinear Programming. *Energy Procedia*, **122**, 1105–1110. [DOI].
- c12 Meggers F., Teitelbaum E., Pantelic J., Rysanek A., **Guo H.** (2017). Liquid Desiccant Latent Load Handling Simulation for Building HVAC Applications with a DOAS Module. *Proceedings of Building Simulation 2017*. [DOI].
- c13 Teitelbaum E., **Guo H.**, Read J., Meggers F. (2017). Mapping Comfort with the SMART (Spherical Motion Average Radiant Temperature) Sensor. *Proceedings of Building Simulation 2017*, **15**. [DOI].
- c14 **Guo H.**, Meggers F. (2015). Impact of Control Availability on Perceived Comfort. *Energy Procedia*, **78**, 1671–1677. [DOI].
- c15 Calabrò E., Meggers F., Teitelbaum E., **Guo H.**, Gmachl C., Penello G.M. (2015). Thermoheliiodome Testing: Evaluation Methods for Testing Directed Radiant Heat Reflection. *Energy Procedia*, **78**, 1762–1768. [DOI].
- c16 Meggers F., Aschwanden G., Teitelbaum E., **Guo H.**, Bruelisauer M. (2015). Urban Cooling Potential: System Losses from Microclimates. *Energy Procedia*, **78**, 3072–3077. [DOI].

Manuscripts under review

- m01 **Guo H.**, He K., Xu Y., Shi Z., Aviv D. (2026). Probabilistic thermal comfort for energy-efficient building control: A risk-aware framework. *Energy Conversion and Management*. Under review.

11 / PATENT APPLICATIONS

- | | |
|------|---|
| 2022 | US 2022/0300860 A1 / Co-inventor
Methods and systems for generating predictions based on time series data using an ensemble model. Patent application published 2022. |
| 2022 | US Provisional 63/300,130 / Co-inventor
Wearable metabolics: Human heat dissipation sensor via non-intrusive core and skin temperature measurement with custom-fitting wearables. Filed 2022. |